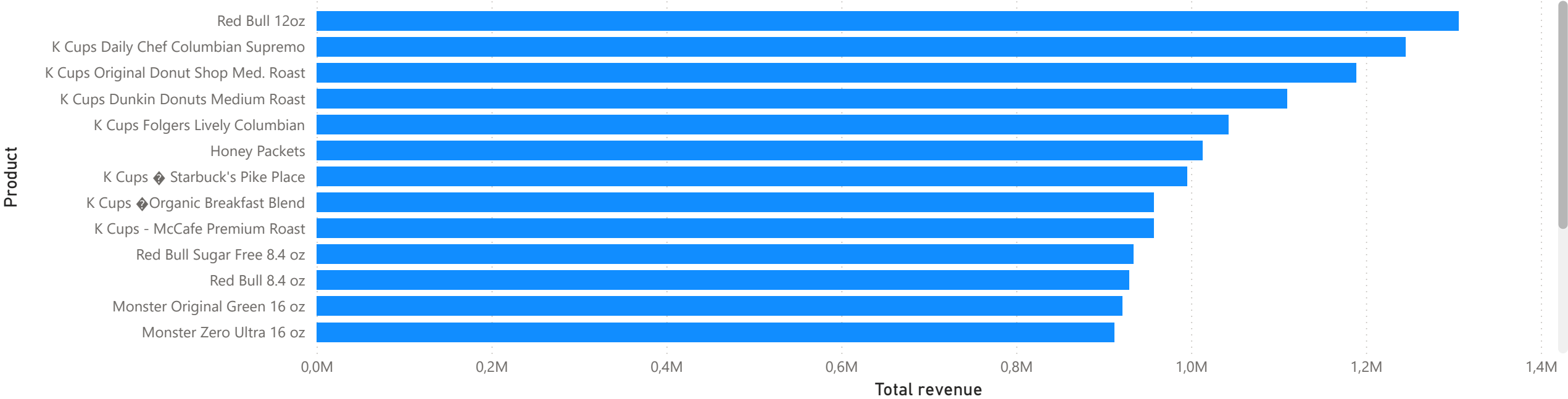


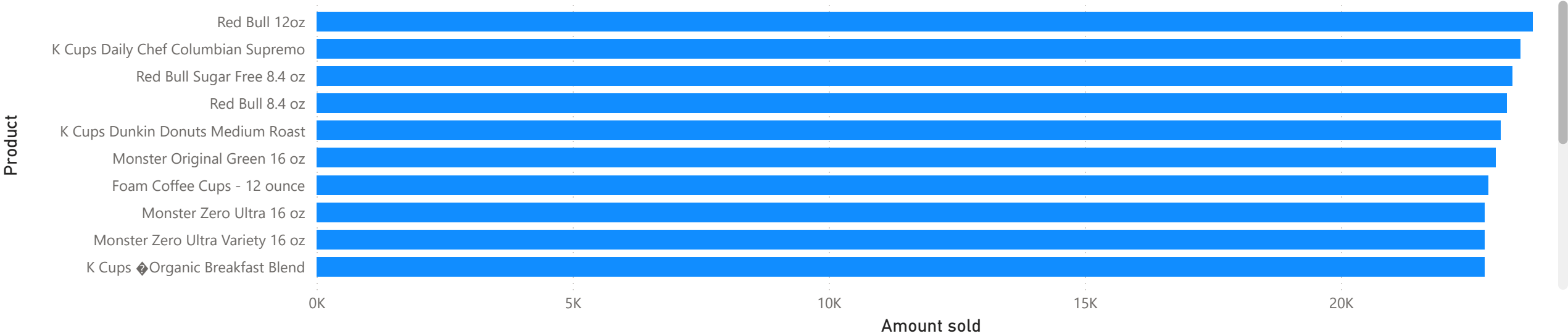
The analysis identified several key findings. **Revenue** is concentrated among a relatively small group of products, customer **retention** remains strong with most customers returning within 30 days, and **sales** show only minor seasonal fluctuations.

Overall, the business demonstrates stable demand driven primarily by repeat customers.

Total revenue by Product

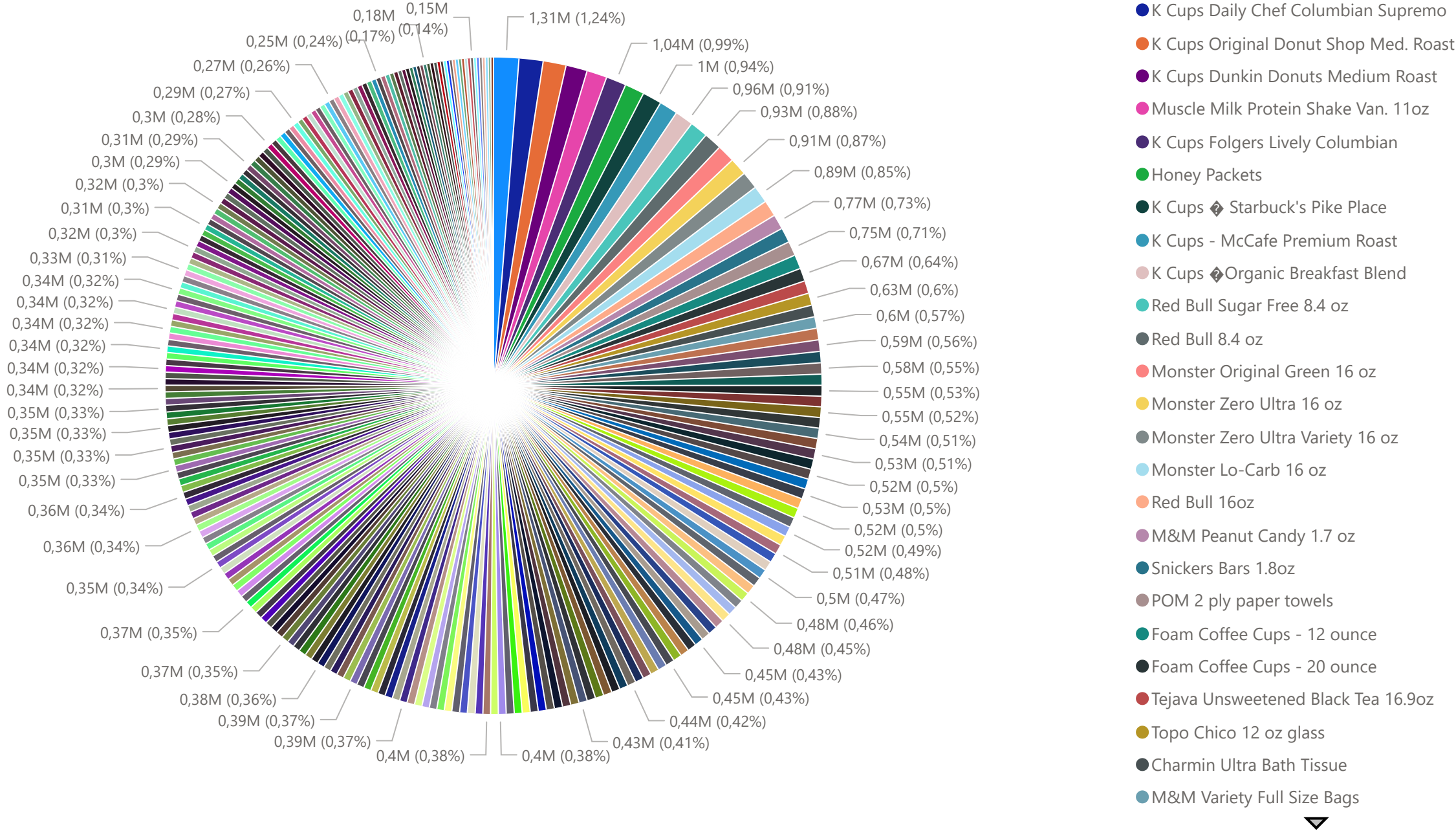


Amount sold by Product

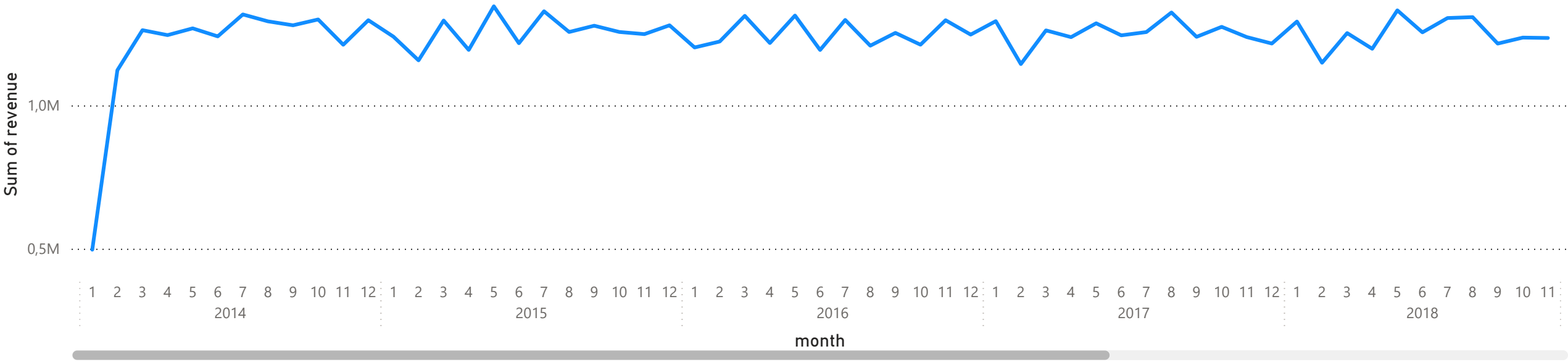


Although unit sales are relatively evenly distributed across products, revenue is heavily skewed towards a small number of high-performing SKUs, suggesting that pricing structure and product value have a stronger impact than volume alone.

Revenue and Revenue share by Product



Sum of revenue by year and month



| month | 2014          | 2015          | 2016          | 2017          | 2018          | 2019          | 2020          | 2021       | Total          |
|-------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|------------|----------------|
| 1     | 496 549,25    | 1 239 765,00  | 1 202 031,50  | 1 293 917,25  | 1 292 509,50  | 1 341 867,00  | 1 291 832,25  | 883 772,25 | 9 042 244,00   |
| 2     | 1 122 547,00  | 1 157 159,75  | 1 222 807,25  | 1 144 129,50  | 1 148 761,50  | 1 149 508,50  | 1 128 964,00  |            | 8 073 877,50   |
| 3     | 1 262 435,75  | 1 296 063,75  | 1 312 405,75  | 1 261 669,25  | 1 252 282,50  | 1 295 730,00  | 1 258 565,00  |            | 8 939 152,00   |
| 4     | 1 245 142,50  | 1 193 564,00  | 1 217 547,00  | 1 238 081,25  | 1 197 510,00  | 1 201 298,00  | 1 211 492,00  |            | 8 504 634,75   |
| 5     | 1 268 770,50  | 1 345 707,75  | 1 313 370,50  | 1 286 428,50  | 1 331 593,25  | 1 266 431,50  | 1 265 700,50  |            | 9 078 002,50   |
| 6     | 1 240 771,75  | 1 216 952,75  | 1 193 281,75  | 1 244 332,00  | 1 254 822,00  | 1 158 766,00  | 1 247 926,75  |            | 8 556 853,00   |
| 7     | 1 317 296,00  | 1 328 373,25  | 1 298 100,75  | 1 255 821,50  | 1 304 771,25  | 1 291 609,75  | 1 250 607,50  |            | 9 046 580,00   |
| 8     | 1 293 211,00  | 1 256 038,75  | 1 208 266,00  | 1 324 086,75  | 1 307 927,75  | 1 257 271,75  | 1 282 673,25  |            | 8 929 475,25   |
| 9     | 1 279 444,25  | 1 277 976,75  | 1 252 952,50  | 1 239 157,00  | 1 215 575,75  | 1 226 580,50  | 1 232 639,75  |            | 8 724 326,50   |
| 10    | 1 299 793,50  | 1 256 143,25  | 1 211 723,50  | 1 274 048,00  | 1 236 509,50  | 1 304 910,25  | 1 319 655,75  |            | 8 902 783,75   |
| 11    | 1 211 569,75  | 1 248 595,75  | 1 297 116,75  | 1 238 362,25  | 1 235 201,25  | 1 248 931,50  | 1 200 248,75  |            | 8 680 026,00   |
| 12    | 1 297 200,00  | 1 279 379,50  | 1 246 905,00  | 1 215 772,75  | 1 330 733,00  | 1 206 605,50  | 1 346 884,75  |            | 8 923 480,50   |
| Total | 14 334 731,25 | 15 095 720,25 | 14 976 508,25 | 15 015 806,00 | 15 108 197,25 | 14 949 510,25 | 15 037 190,25 | 883 772,25 | 105 401 435,75 |

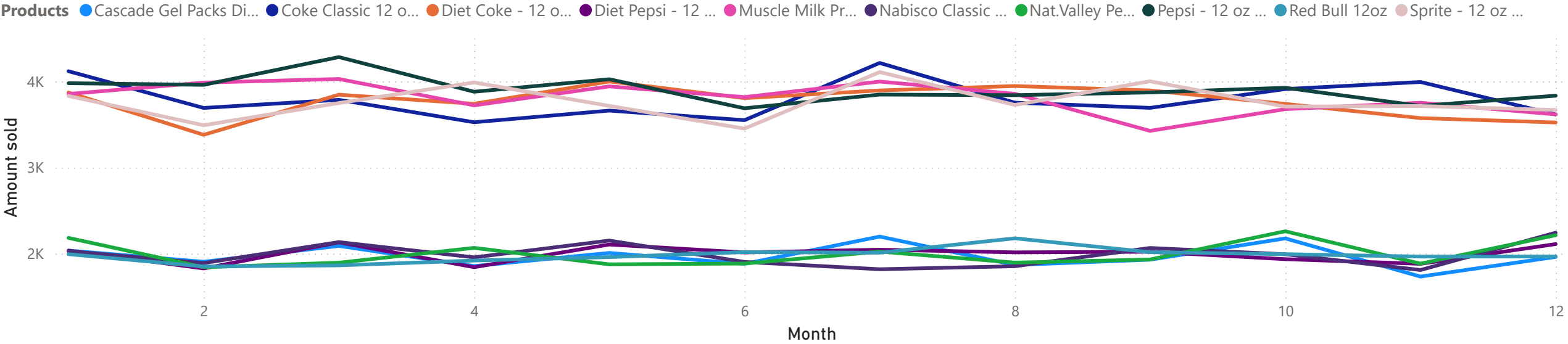
Revenue remains relatively stable over the observed period, fluctuating around an average monthly level of approximately 1.2–1.3 million. The heatmap and line chart together indicate no strong long-term upward or downward trend, suggesting a generally stable demand structure over time.

A recurring drop in revenue is consistently observed in February across multiple years, which may indicate a seasonal pattern or post-holiday demand normalization. In contrast, April shows slightly lower performance compared to other months, although this effect is less consistent.

Overall, the dataset shows weak and inconsistent seasonality, with February being the only clearly recurring low point, while the remaining fluctuations appear largely driven by noise rather than systematic patterns.

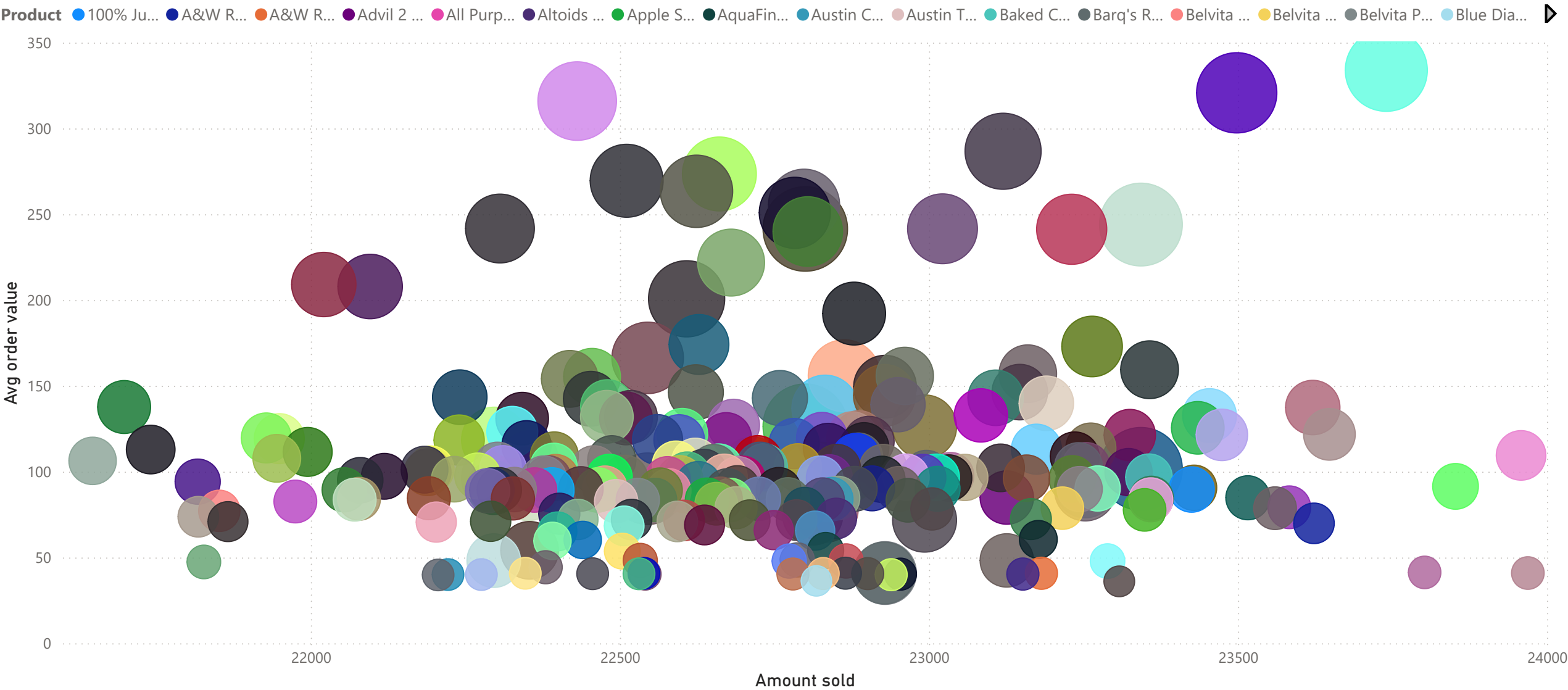
| Product                                | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      | 9      | 10     | 11     | 12     | Total   |
|----------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| Pepsi - 12 oz cans                     | 3978   | 3959   | 4281   | 3879   | 4024   | 3687   | 3846   | 3839   | 3871   | 3925   | 3715   | 3833   | 46837   |
| Muscle Milk Protein Shake Van. 11oz    | 3853   | 3985   | 4027   | 3721   | 3941   | 3817   | 3997   | 3855   | 3425   | 3677   | 3751   | 3616   | 45665   |
| Coke Classic 12 oz cans                | 4117   | 3691   | 3785   | 3525   | 3660   | 3549   | 4213   | 3752   | 3692   | 3910   | 3992   | 3615   | 45501   |
| Diet Coke - 12 oz cans                 | 3870   | 3379   | 3845   | 3739   | 3998   | 3805   | 3894   | 3945   | 3894   | 3738   | 3573   | 3522   | 45202   |
| Sprite - 12 oz cans                    | 3830   | 3490   | 3748   | 3986   | 3715   | 3451   | 4107   | 3726   | 4000   | 3707   | 3713   | 3667   | 45140   |
| Diet Pepsi - 12 oz cans                | 2035   | 1828   | 2131   | 1843   | 2106   | 2014   | 2046   | 2015   | 2022   | 1936   | 1882   | 2111   | 23969   |
| Nat.Valley PeanutButter Protein Bars   | 2183   | 1842   | 1895   | 2067   | 1876   | 1885   | 2024   | 1896   | 1932   | 2260   | 1884   | 2214   | 23958   |
| Nabisco Classic Mix ccooki, cracker    | 2029   | 1891   | 2134   | 1957   | 2152   | 1904   | 1819   | 1854   | 2066   | 1993   | 1810   | 2243   | 23852   |
| Red Bull 12oz                          | 1994   | 1851   | 1864   | 1920   | 1960   | 2018   | 2011   | 2178   | 2017   | 1994   | 1966   | 1967   | 23740   |
| Cascade Gel Packs Dishwasher           | 2036   | 1906   | 2090   | 1856   | 2008   | 1882   | 2198   | 1869   | 1930   | 2177   | 1734   | 1962   | 23648   |
| A&W Root Beer - 12 oz cans             | 2046   | 1848   | 2071   | 1948   | 1783   | 2004   | 2136   | 2098   | 1869   | 2014   | 1977   | 1829   | 23623   |
| Starbuck's Pike Place                  | 2093   | 1799   | 2189   | 1974   | 2278   | 1848   | 1904   | 1967   | 2075   | 1890   | 1763   | 1841   | 23621   |
| Kellogg's Cereals Assortment Pack      | 2112   | 1875   | 2042   | 1798   | 1855   | 1884   | 1926   | 2039   | 2021   | 1995   | 1726   | 2310   | 23583   |
| Quaker Instant Oatmeal Maple Brn Sugar | 2232   | 1746   | 1790   | 1582   | 2167   | 1907   | 2029   | 1977   | 2004   | 1995   | 2120   | 2010   | 23560   |
| Total                                  | 515352 | 459295 | 509009 | 486156 | 517258 | 485654 | 516339 | 509136 | 496247 | 507718 | 491338 | 506683 | 6000185 |

Amount sold by Month and Products



- Key findings
- Beverage products dominate sales volume, with Pepsi, Protein Shake, Coke, Diet Coke and Sprite consistently ranking among the best-selling items.
  - Monthly sales remain relatively stable throughout the observed period, indicating steady customer demand rather than strong seasonality.
  - A small group of products accounts for a disproportionately large share of total units sold, suggesting that business performance is heavily driven by a limited number of high-performing SKUs.

Sum of revenue by Product, Amount sold and Avg order value



**Product Revenue Structure**

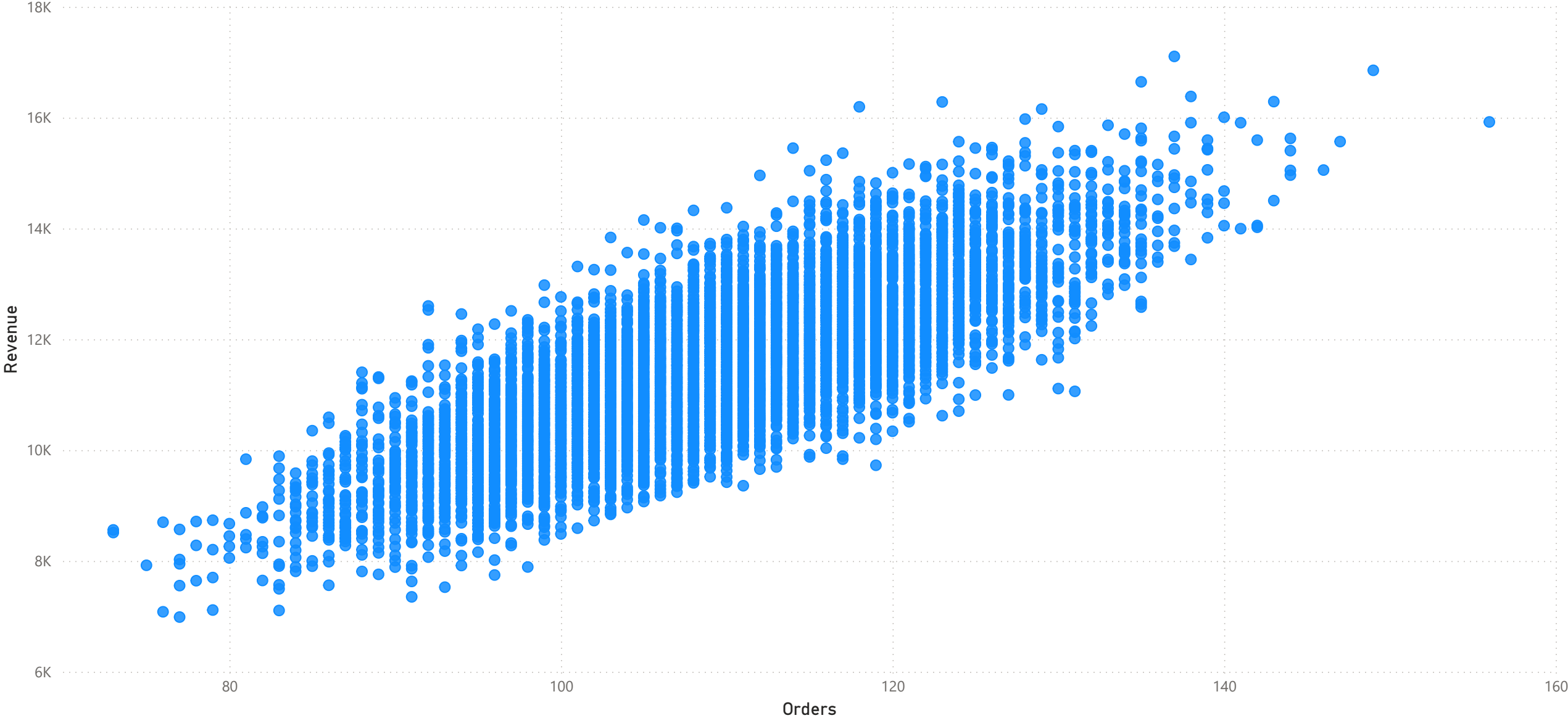
The scatter plot shows a highly concentrated distribution of products in terms of units sold, with most items clustered within a narrow range of approximately 22,000–24,000 units. This suggests a relatively balanced demand across the majority of the product portfolio.

Revenue differences are primarily driven by variations in average order value rather than sales volume. While most products generate moderate revenue levels, a small number of high-performing items (e.g., Red Bull and coffee products) significantly outperform others in both revenue contribution and transaction value.

Overall, the business does not appear to be dominated by a single product in terms of volume, but rather by a small set of high-value products that disproportionately contribute to total revenue.



Orders and Revenue by customer

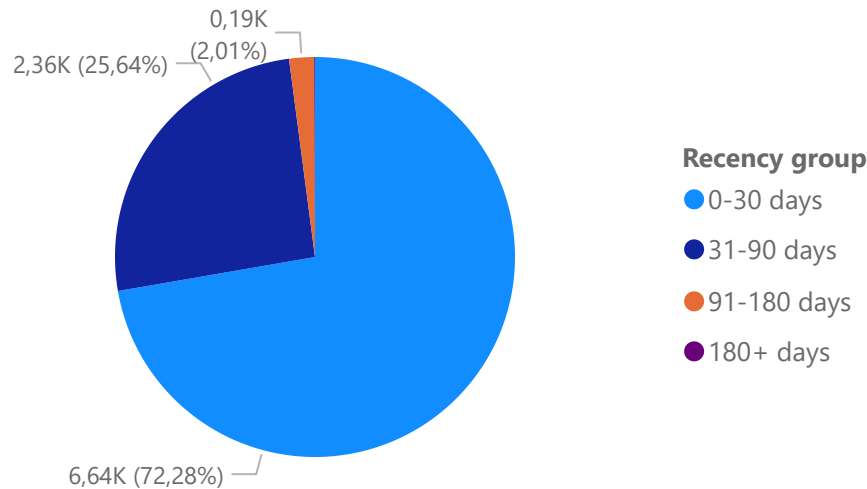


**Customer Frequency vs Monetary Value**

The scatter plot shows a clear positive relationship between purchase frequency and total customer revenue. Customers who place more orders generally generate higher revenue, indicating a strong association between customer activity and business value.

Most customers follow a consistent spending pattern with relatively few outliers, suggesting stable purchasing behavior across the customer base. The distribution also indicates a substantial number of repeat customers rather than predominantly one-time buyers.

Customer number by Recency group



105,40M

Total revenue

9191

Customers amount

1,00M

Count of transactions

105,40

Average order price

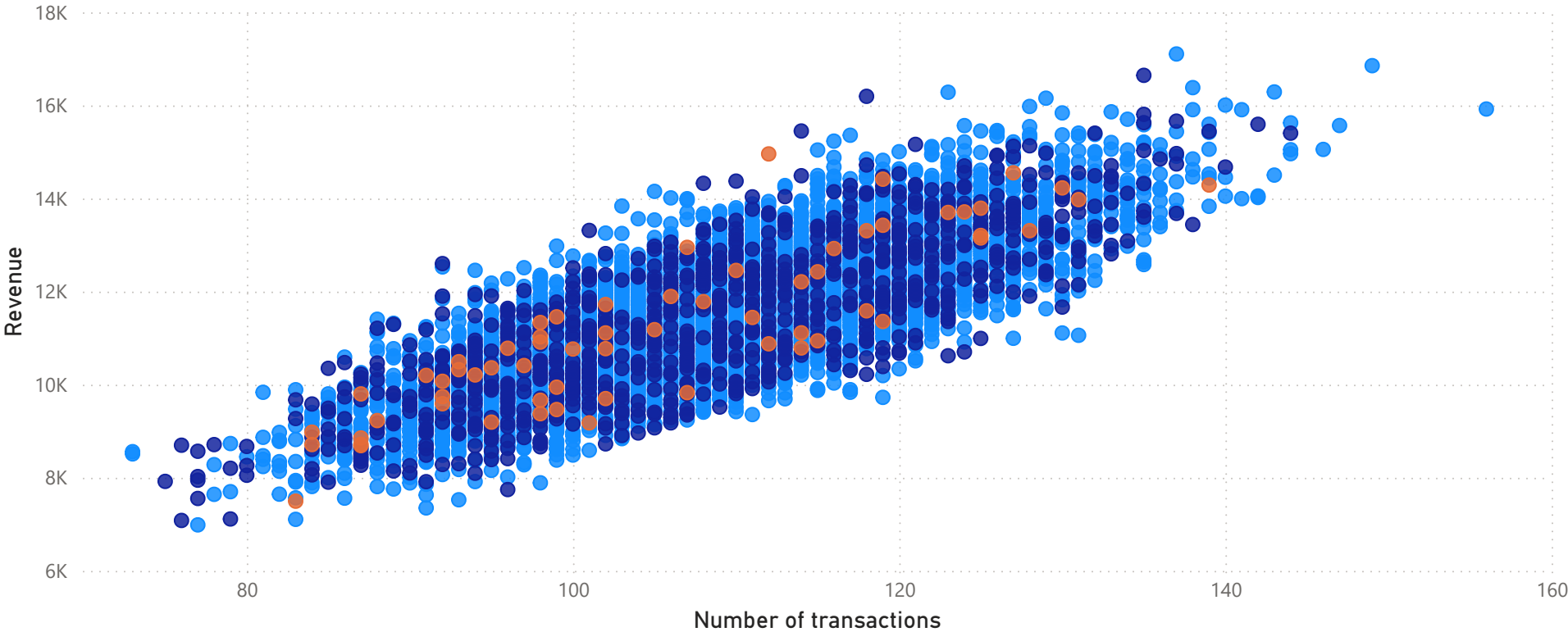
The customer base is predominantly active, with 72% of customers classified as regular purchasers. A further 25% show moderate activity, indicating occasional engagement rather than full churn. Only a small fraction of customers fall into low-activity or inactive segments, suggesting strong overall retention and a stable returning customer base. This indicates that the business is primarily driven by repeat customers rather than one-time purchases, with churn levels remaining relatively low.



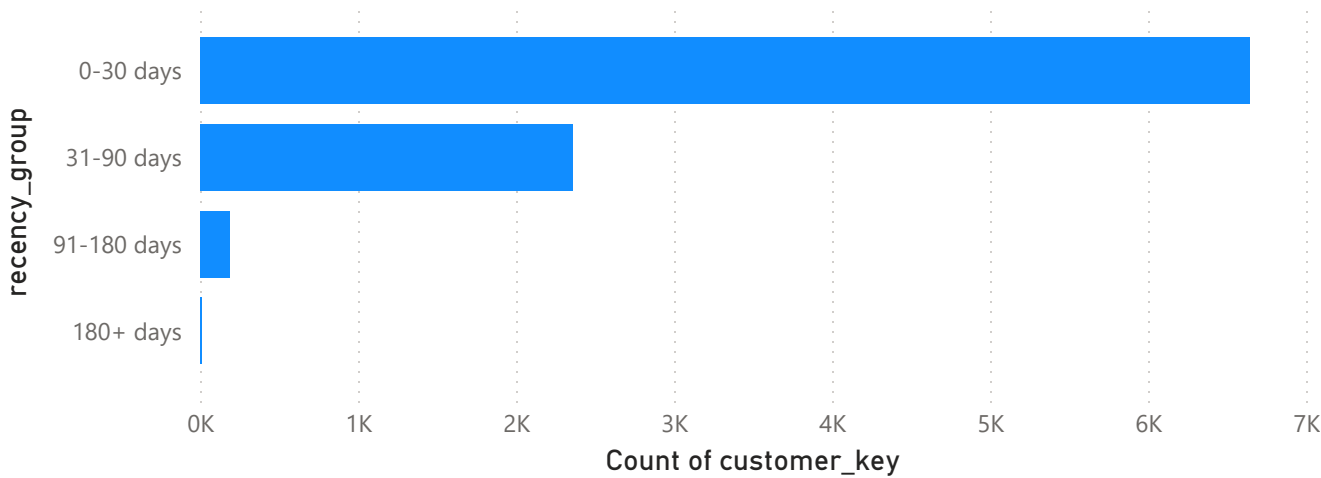
| customer_key | recency | frequency | monetary  |
|--------------|---------|-----------|-----------|
| C004349      | 9       | 137       | 17 104,50 |
| C005316      | 9       | 149       | 16 853,25 |
| C000273      | 64      | 135       | 16 645,75 |
| C001438      | 7       | 138       | 16 381,50 |
| C007553      | 13      | 143       | 16 288,75 |
| C007415      | 11      | 123       | 16 282,50 |
| C008009      | 73      | 118       | 16 194,75 |
| C007205      | 24      | 129       | 16 154,50 |
| C002968      | 11      | 140       | 16 006,50 |
| C006902      | 18      | 128       | 15 977,25 |
| C007500      | 26      | 128       | 15 973,75 |
| C006906      | 19      | 156       | 15 922,00 |
| C002670      | 20      | 138       | 15 908,00 |
| C001677      | 0       | 141       | 15 907,25 |
| C005345      | 1       | 133       | 15 862,00 |
| C006350      | 1       | 130       | 15 838,00 |
| C005960      | 30      | 135       | 15 807,25 |
| C009168      | 1       | 134       | 15 702,50 |
| C006326      | 33      | 137       | 15 661,25 |
| C004990      | 42      | 135       | 15 628,00 |
| C000152      | 7       | 144       | 15 624,50 |
| C006083      | 8       | 139       | 15 594,25 |
| C003577      | 83      | 142       | 15 593,75 |
| C007924      | 8       | 135       | 15 584,00 |
| C008798      | 6       | 147       | 15 567,50 |
| C006717      | 1       | 124       | 15 565,50 |
| C001197      | 15      | 128       | 15 544,50 |
| C000590      | 23      | 126       | 15 459,50 |
| C007549      | 3       | 125       | 15 450,25 |
| C000603      | 63      | 114       | 15 449,25 |
| C003115      | 34      | 139       | 15 442,25 |
| C006412      | 10      | 137       | 15 436,00 |
| C002645      | 26      | 126       | 15 421,50 |
| C007163      | 12      | 139       | 15 420,50 |

Recency group, Number of transactions and Revenue

Recency group 0-30 days 31-90 days 91-180 days



Count of customer\_key by recency\_group



Customer behavior analysis indicates that the business is supported by a strong base of active repeat customers. Most customers have made purchases within the last 30 days, while only a very small proportion has been inactive for extended periods. The positive relationship between purchase frequency and monetary value suggests that encouraging repeat purchases is a key driver of long-term revenue growth.